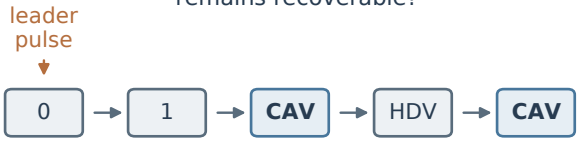
(a) Finite-amplitude recovery

What requested disturbance amplitude remains recoverable?

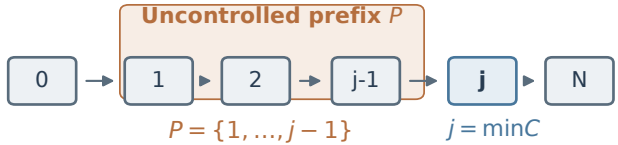


Directed chain; sparse active vehicles

Path: $s_i(t) \geq s_{\min}$
Terminal: $v_i \to \bar{v}$, $s_i \to \bar{s}_i$, $a_i \to 0$
Loss: $\bar{J} \leq \ell$ Deadline: T_r

System-level limit $\alpha_\star$

(b) Directed prefix obstruction



$s_i < s_{\min}$
Required safety-bound violation

Downstream actuation cannot change this prefix trajectory
Fixed leader path, delayed histories, parameters, active set and directed topology

Prefix safety violation → conditional policy-independent obstruction

(c) Complementary evidence

NONLINEAR
SAME-MODEL WITNESS

Fixed leader path/history

Heterogeneous nonlinear dynamics

Logged positive-gap prefix crossing

Policy-independent
obstruction (fixed run)

SAMPLED-LINEAR
RESERVE

$$Q = Q_0 + \alpha L_\pi d$$

$$m_r = b_r - c_r^T Q_0$$

$$\gamma_r = h_D(L_\pi^T c_r)$$

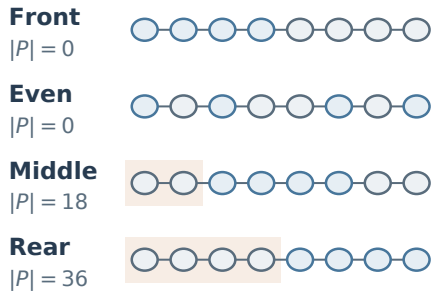
$$\alpha_{\pi,\,aff} = \min\{\alpha_{max}, \min_{r:\,\gamma_r > 0} m_r/\gamma_r\}$$

Limiting row:
vehicle × time × constraint
Conditional upper bound

Distinct nonlinear and sampled-linear evidence

(d) Spatial implication

Four active CAVs in each nonlinear $N = 40$ layout
Schematic; not to scale



Equal tail transfer can coexist with different local constraint reserves (Supplementary Fig. S2).